

Lagrangian Information Geometric Regularization



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Compressible Euler Equations

(With IGR pressure)

$$\partial_t \begin{pmatrix} \rho \\ \rho \mathbf{u} \\ E \end{pmatrix} + \nabla \cdot \begin{pmatrix} \rho \mathbf{u} \\ \rho \mathbf{u} \mathbf{u}^T + p I \\ (E + p) \mathbf{u} \end{pmatrix} = 0$$

$$p \mapsto p + \Sigma$$

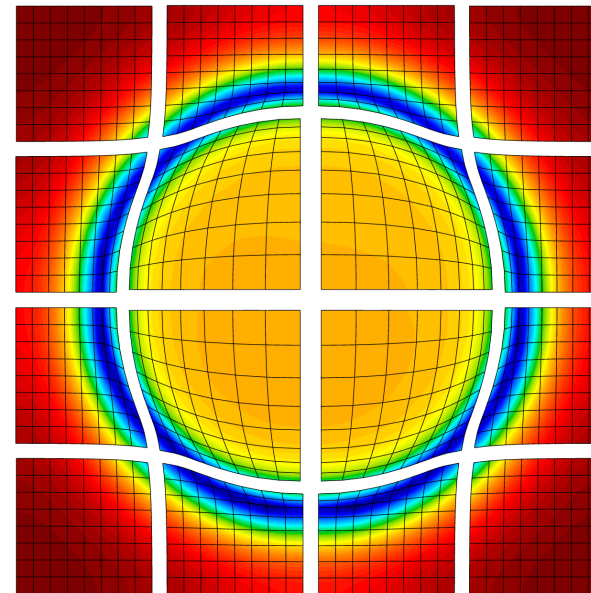
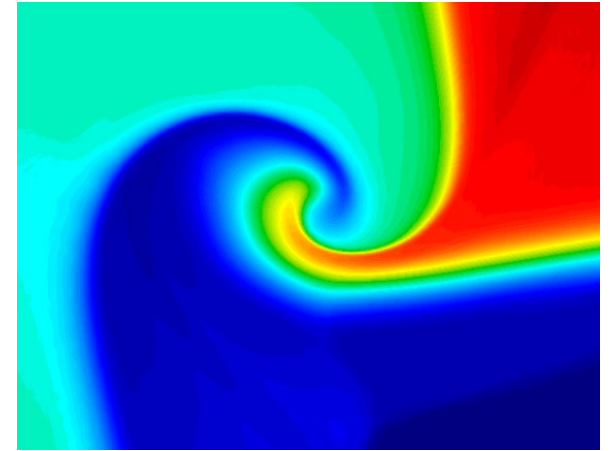
$$\rho \frac{dv}{dt} = \nabla \cdot \sigma, \quad \frac{1}{\rho} \frac{d\rho}{dt} = -\nabla \cdot v, \quad \rho \frac{de}{dt} = \sigma : \nabla v, \quad \frac{dx}{dt} = v, \quad \sigma = -\text{EOS}(\rho, e) I$$

$$E = \rho e + \frac{1}{2} \rho |\mathbf{u}|^2, \quad \mathbf{v} \equiv \mathbf{u}$$

$$\sigma \mapsto \sigma + \Sigma I$$

Benefits of IGR

- Resolve macro behavior at coarser mesh scales
- Larger Time Steps
- Easy to implement
- Greatly parallelizable



LAGrangian High-Order Solver (LAGHOS)

- C++ MFEM based library on curvilinear meshes
- Solves the compressible Euler equations in multiple dimensions
- Kinematic Variables (position, velocity) are in H^1 i.e. continuous.
- Thermodynamic variables (pressure, energy) are in L^2 i.e discontinuous.
- Visualization in ParaView and Glvis
- Parallelizable with OpenMPI and CUDA
- Partial Assembly to speedup and reduce storage

IGR

- Instead of using viscosity, we modify the stress tensor.

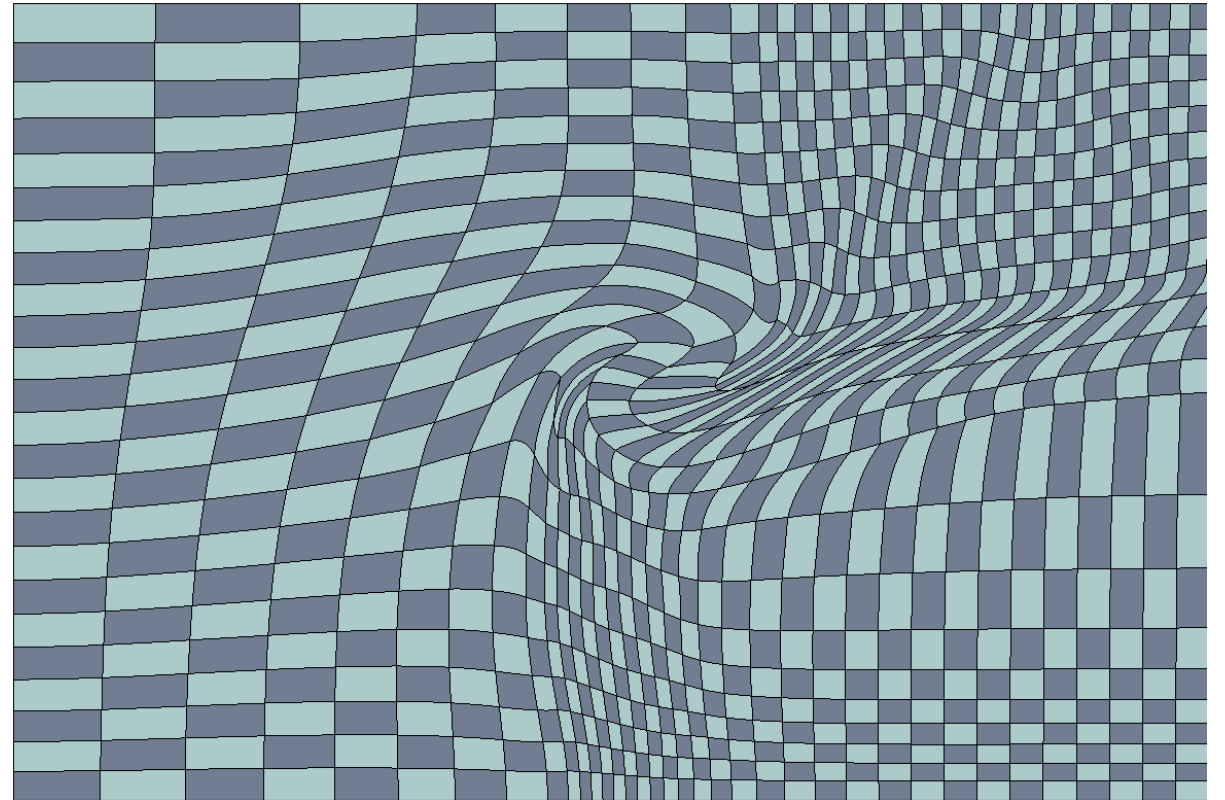
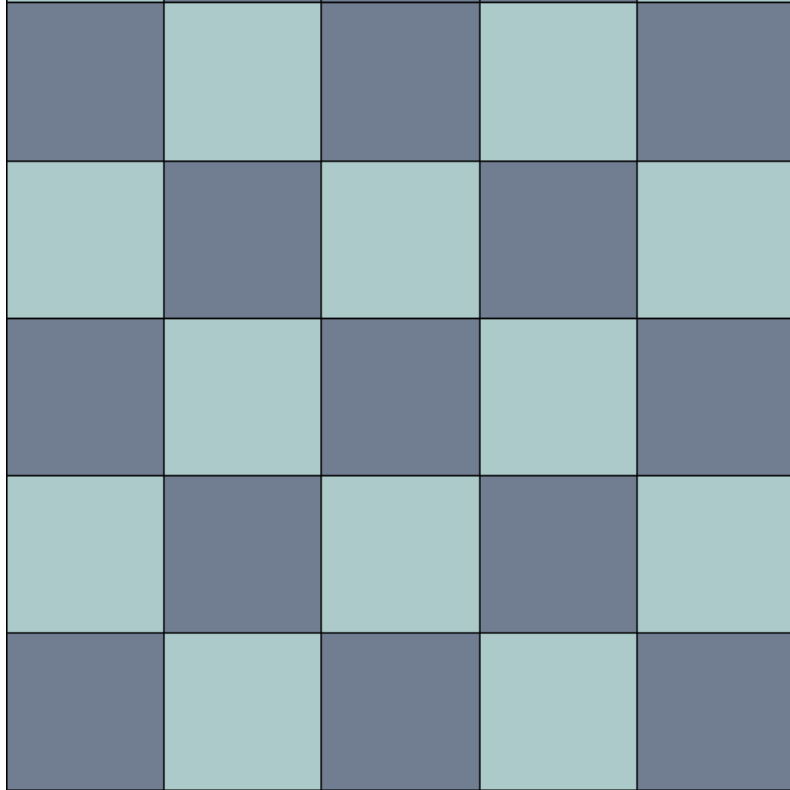
$$\boxed{p \mapsto p + \Sigma} \quad \text{or} \quad \boxed{\sigma \mapsto \sigma + \Sigma I}$$

- This new term, the igr pressure costs an extra elliptic solve per RK step.

$$\boxed{\rho^{-1} \Sigma - \alpha \operatorname{div} (\rho^{-1} \nabla \Sigma) = \alpha (\operatorname{tr}([D\mathbf{u}])^2 + \operatorname{tr}([D\mathbf{u}]^2))}$$

- We take Σ to be in H^1 of the same order as the kinematic variables.
- Laghos time stepping continues as before.

Eulerian vs Lagrangian Coordinates

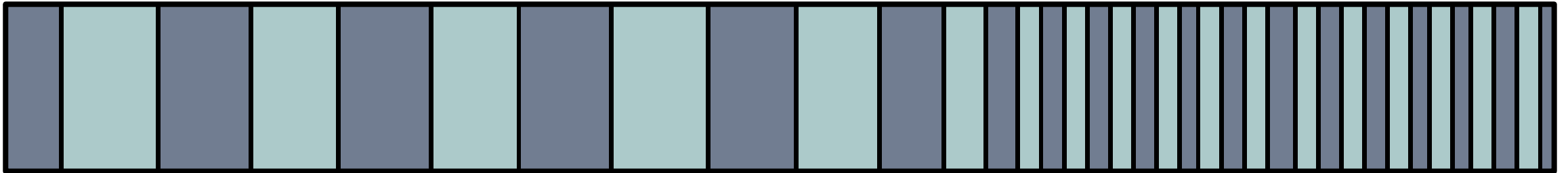


- The mesh is curvilinear.

What value of Alpha?

Eulerian coordinates: $\alpha \simeq C dx^2$ with $C \sim 1 - 10$.

Lagrangian coordinates: dx changes over each element and in time.



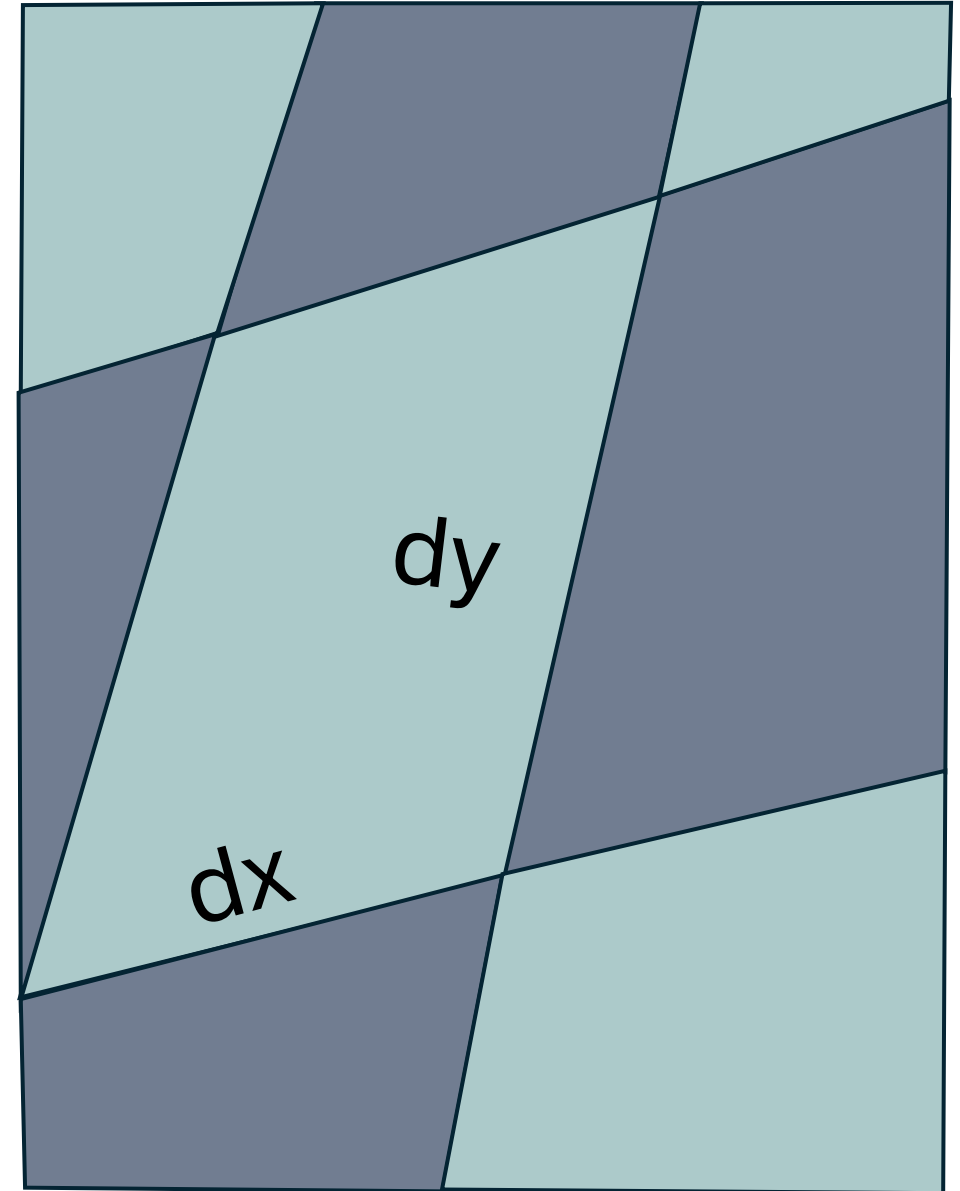
Fixed Alpha:

- Too much regularization near shocks
- Not enough regularization away from shocks

Alpha in 2d

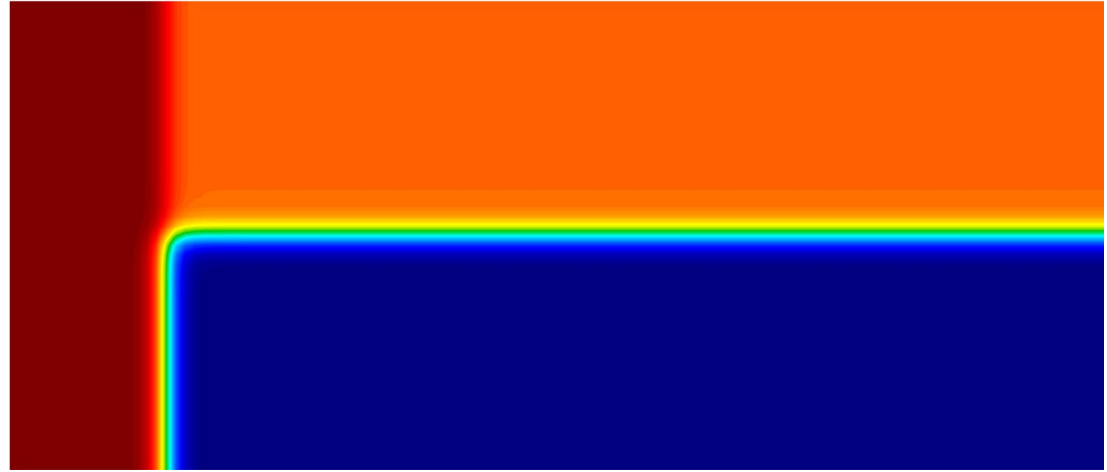
Multiple Choices:

- $\alpha = C \sqrt{dx \, dy}$
- $\alpha = C \max \{ dx, dy \}$
- $\alpha = C F^\top F$ (Cauchy-Green)



Smoothing

- For now, we add some initial smoothing to the initial conditions.



Smooth Triple Point IC



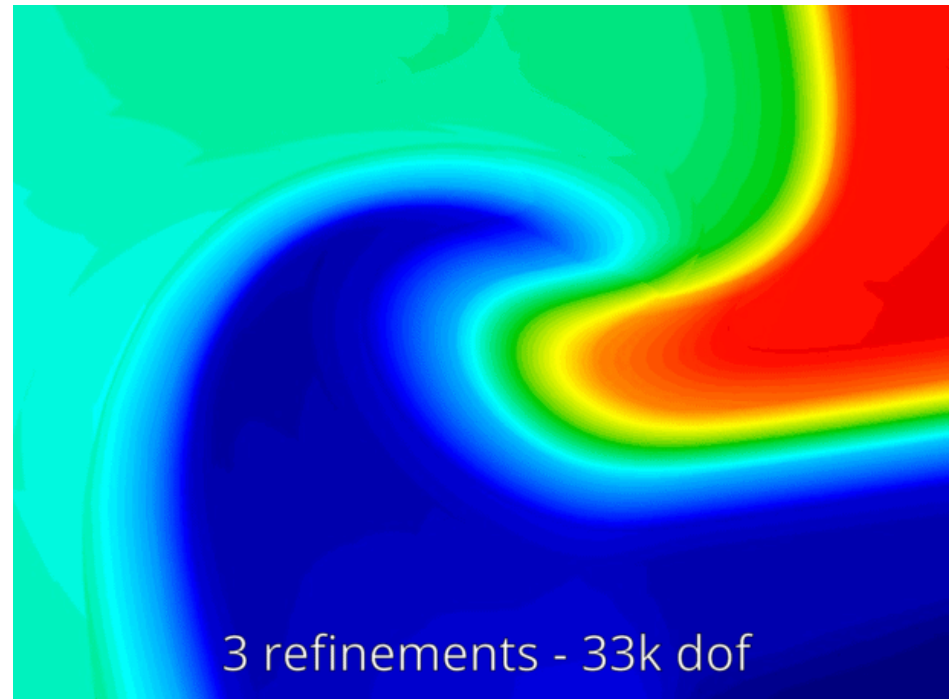
Sharp Triple Point IC

Triple Point Gif

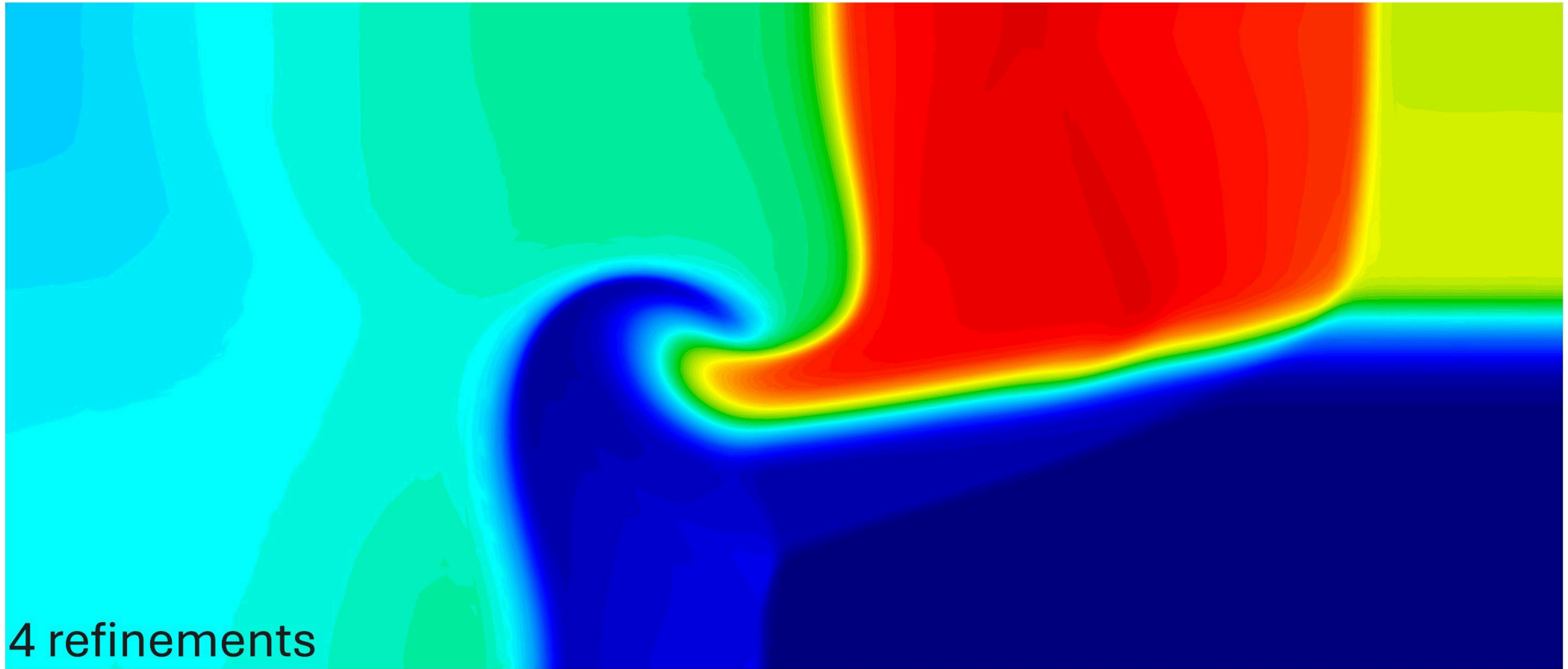


Refinement changes Macro Behavior

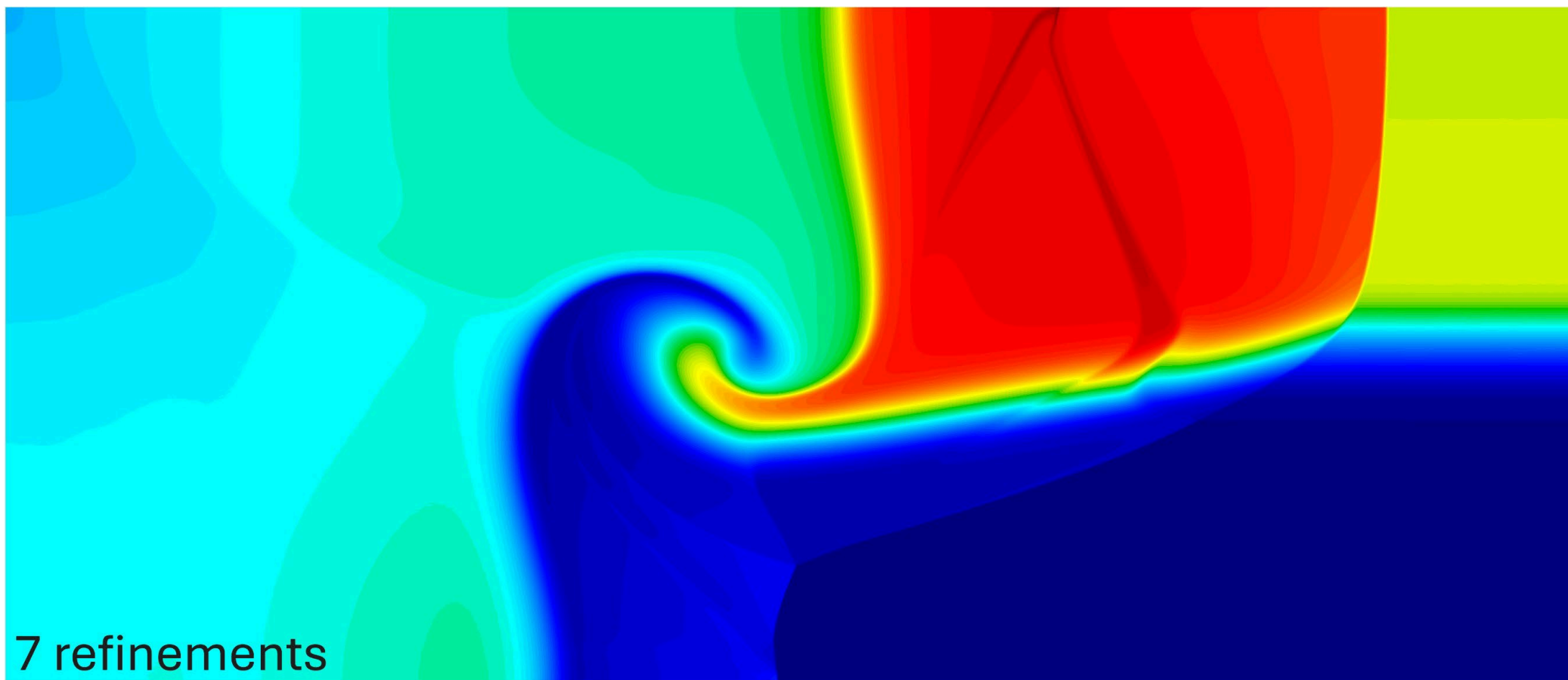
- More refinements allows for more twisting



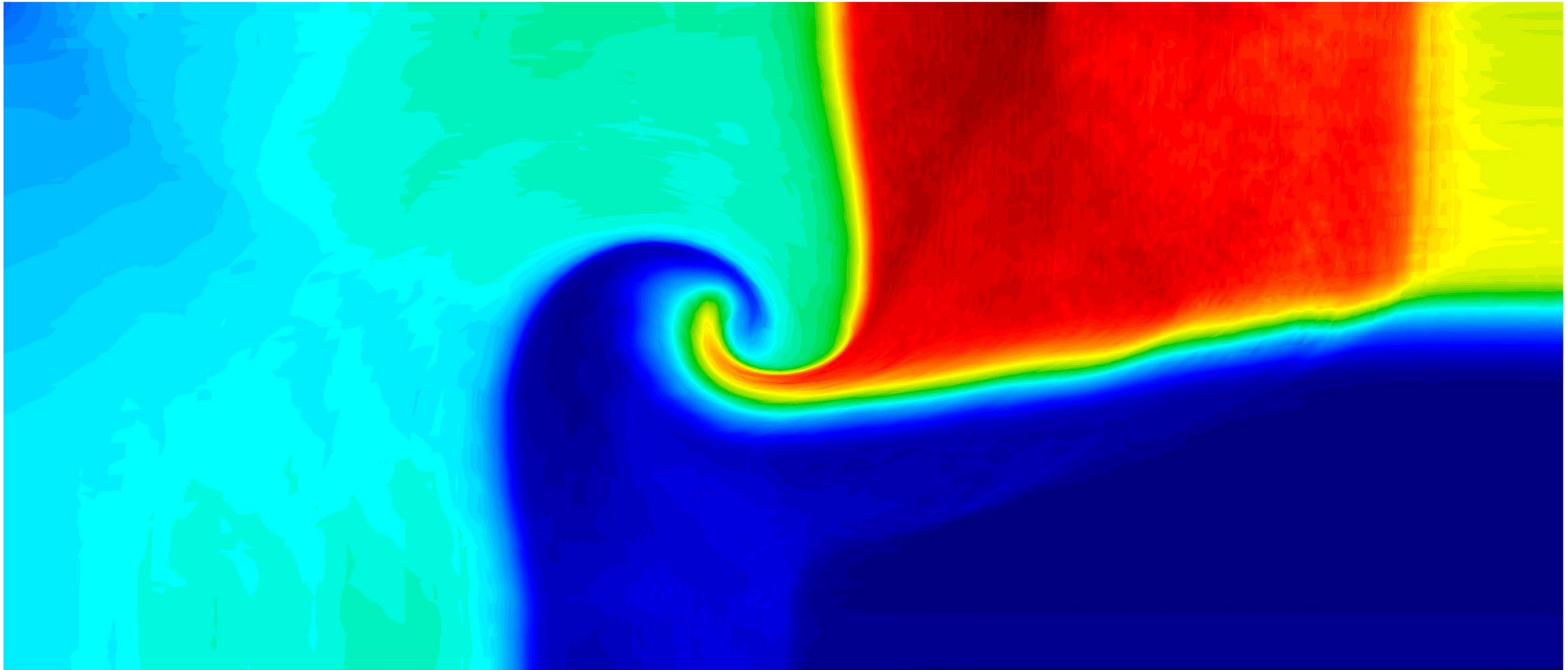
4 refinements



7 refinements

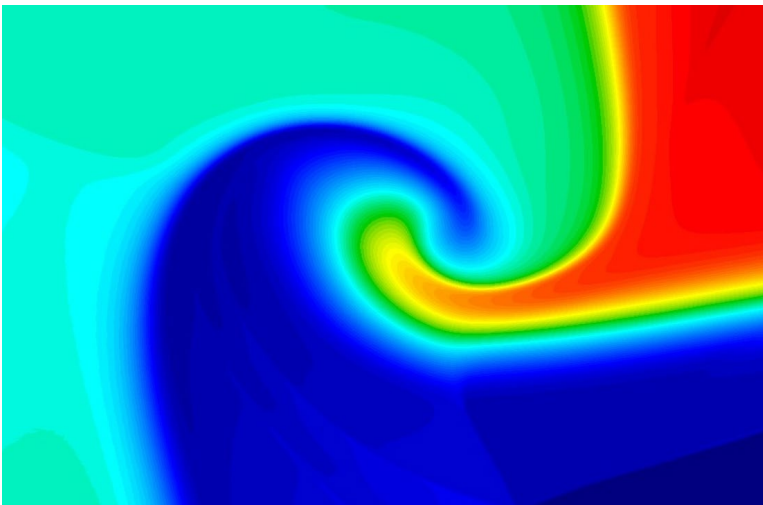


With IGR (4 refinements)

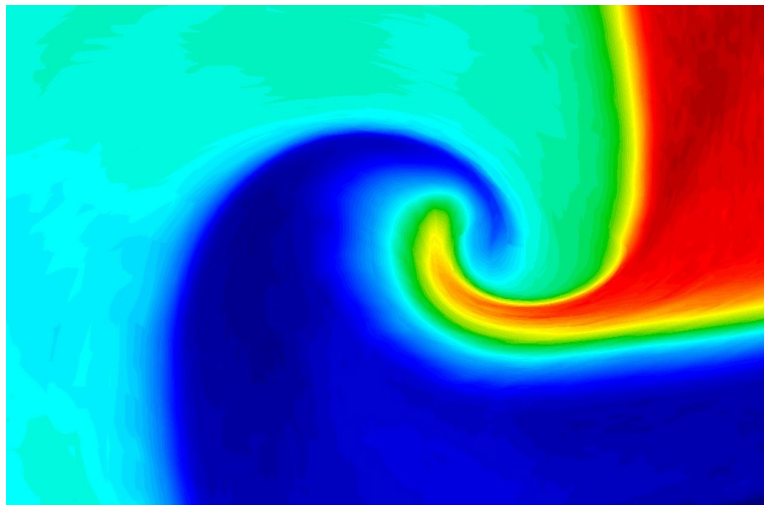


How does IGR Compare?

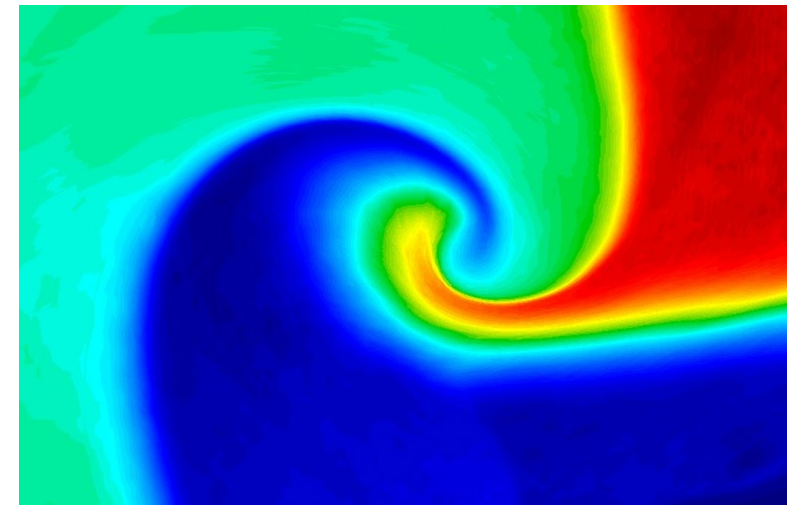
- IGR outperforms pure Laghos with much lower refinements
- Times given per Perlmutter node-min (128 cores)



Pure Laghos, 7 refinements
8.25M dof
5380 node-min



Laghos + IGR, 4 refinements
130k dof
81 node-min



Laghos + IGR, 5 refinements
520k dof
155 node-min

More Runtime Information

- IGR can speed up runs by:
 - Requiring lower resolution
 - Improving CFL number.

```
step 20040,    t = 2.9991,    dt = 0.000047,    |e| = 2.1709576864e+02
step 20050,    t = 2.9996,    dt = 0.000047,    |e| = 2.1709914956e+02
step 20059,    t = 3.0000,    dt = 0.000032,    |e| = 2.1710208076e+02
results/WithoutIGR/p15/Laghos_15_4_21313_3e+03_noigr_LaghosVisc
Execution time: 541.3035043220 seconds.

CG (H1) total time: 84.0500481140
CG (H1) rate (megadofs x cg_iterations / second): 565.3588387665

CG (L2) total time: 4.0138568030
CG (L2) rate (megadofs x cg_iterations / second): 80.7634193023

Forces total time: 99.8144403700
Forces rate (megadofs x timesteps / second): 39.3608999403

UpdateQuadData total time: 268.0478232330
UpdateQuadData rate (megaquads x timesteps / second): 19.3636247793

IGR Pressure total time: 0.0000000000

Major kernels total time (seconds): 445.0857188870
Major kernels total rate (megadofs x time steps / second): 127.2509965353

Energy diff: 2.49e-05
dave@DESKTOP-D56MBPN:/mnt/c/Users/Dave/LaghosScal4/laghos$
```

```
step 1930,    t = 2.9904,    dt = 0.000494,    |e| = 2.2231936434e+02
step 1940,    t = 2.9954,    dt = 0.000494,    |e| = 2.2236058575e+02
step 1950,    t = 3.0000,    dt = 0.000172,    |e| = 2.2239855676e+02
results/WithIGR/alpha=4000.00e-3/p15/Laghos_15_4_21333_3e+03_0.00
Execution time: 127.4909935760 seconds.

CG (H1) total time: 10.6226928950
CG (H1) rate (megadofs x cg_iterations / second): 500.5081717558

CG (L2) total time: 0.3877440870
CG (L2) rate (megadofs x cg_iterations / second): 81.9410561379

Forces total time: 9.1237510420
Forces rate (megadofs x timesteps / second): 42.2041480776

UpdateQuadData total time: 32.0332798710
UpdateQuadData rate (megaquads x timesteps / second): 15.9259650605

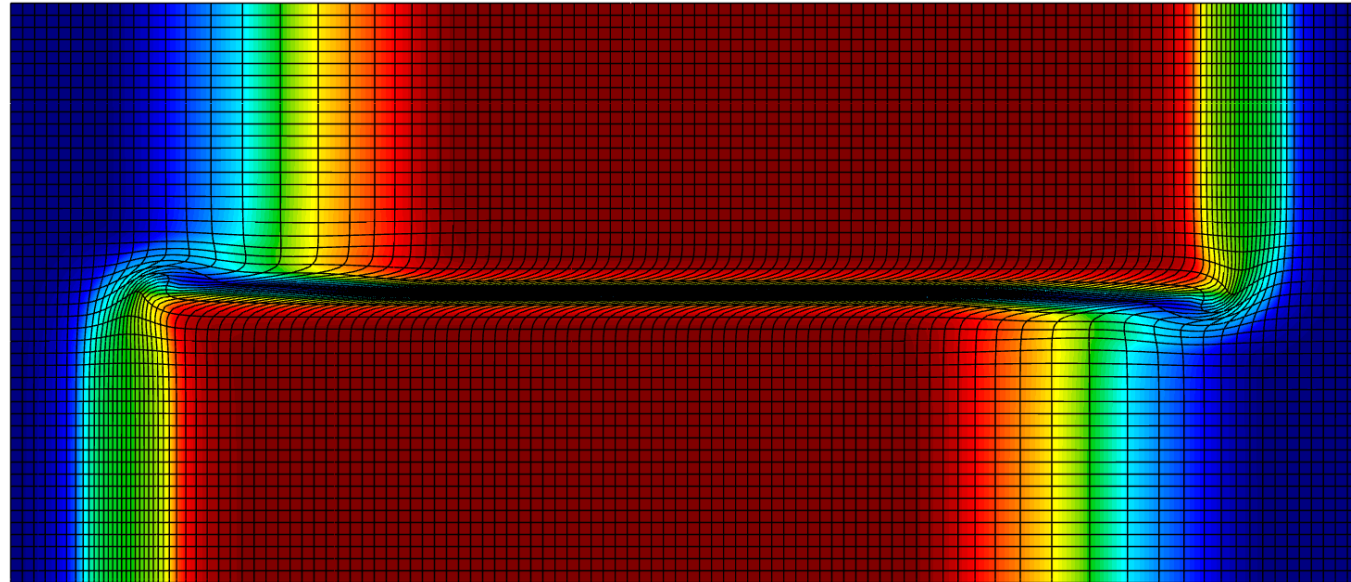
IGR Pressure total time: 39.0853544820

Major kernels total time (seconds): 51.6851172690
Major kernels total rate (megadofs x time steps / second): 120.1886725664
```

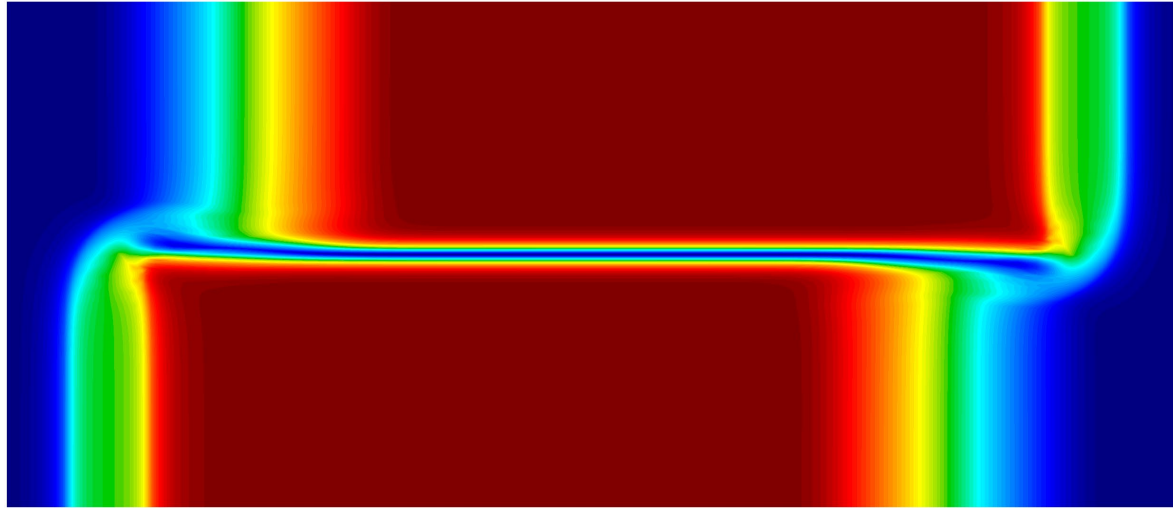
IGR has 10x fewer time steps for same number of refinements.

Pure Shearing Run

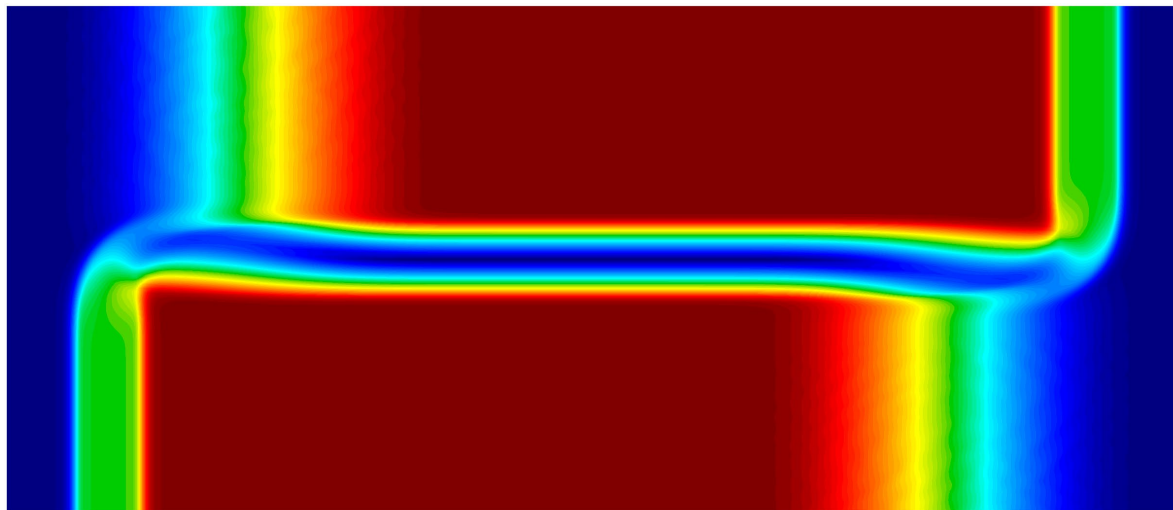
- IGR speeds things up considerably in shearing simulations.
- With IGR it is 38 sec vs 638 seconds for Laghos (16x speedup).



Artificial Viscosity Widens Shearing Layers



IGR no
Artificial
Viscosity



Pure
Laghos

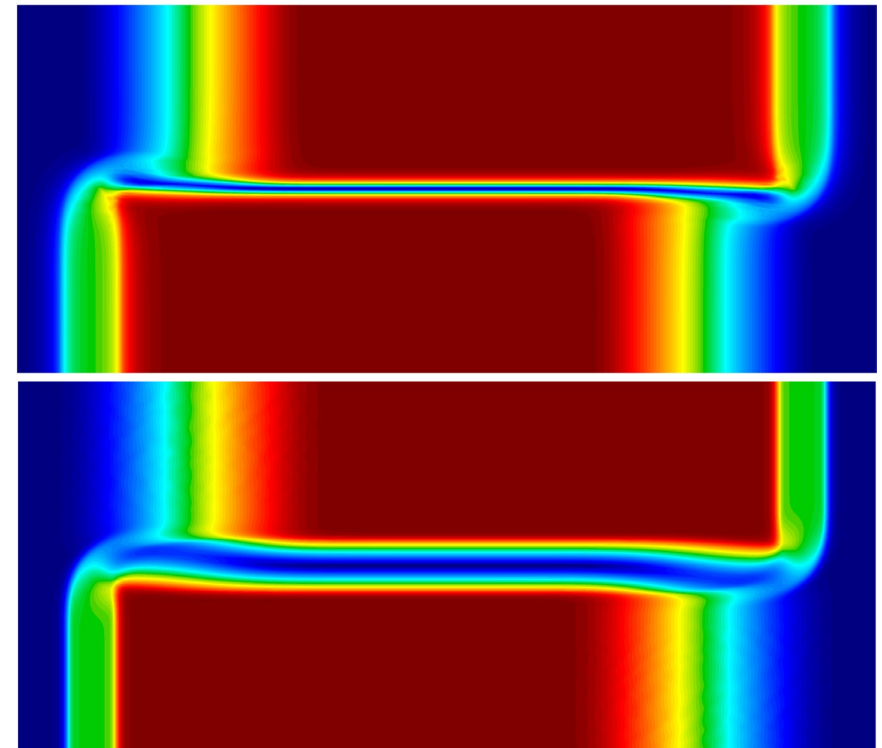
Why does IGR improve shearing runs?

- Larger Time Steps $\Delta t_q = \frac{CFL}{c_s/h_{min} + 2.5\mu_{visc}/\rho h_{min}^2}$
- Artificial Viscosity slows down shearing

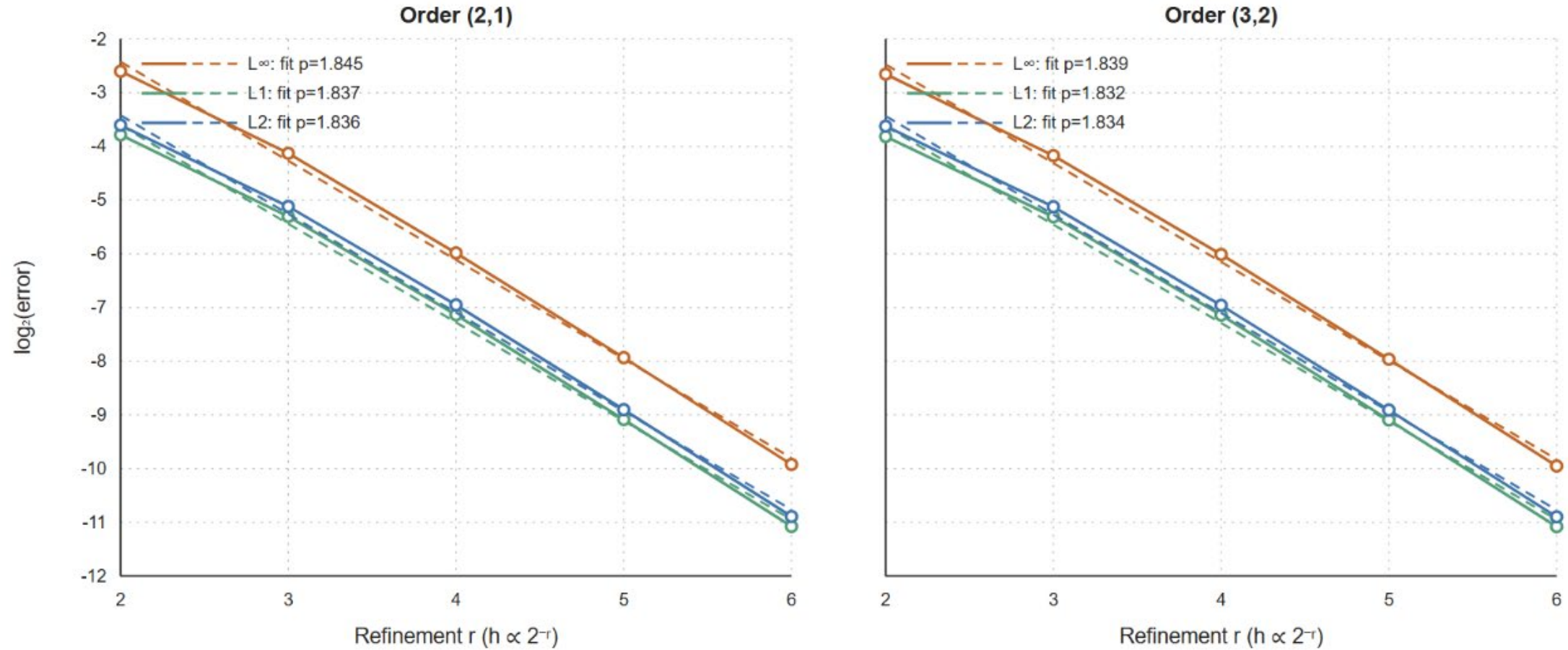
$$\rho \frac{du}{dt} = \nabla \cdot \sigma \quad \sigma_{visc} = \frac{1}{2} \mu_s(x) (\nabla u + \nabla u^T)$$

$$\left. \frac{du}{dt} \right|_{(x,t)=(0,0)} = \begin{pmatrix} -396y - 79y\psi_0 + O(y^3) \\ 0 \end{pmatrix}$$

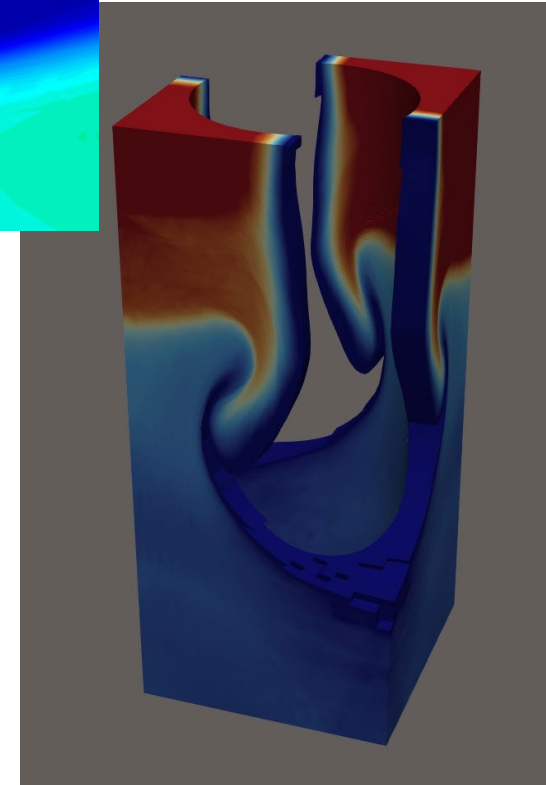
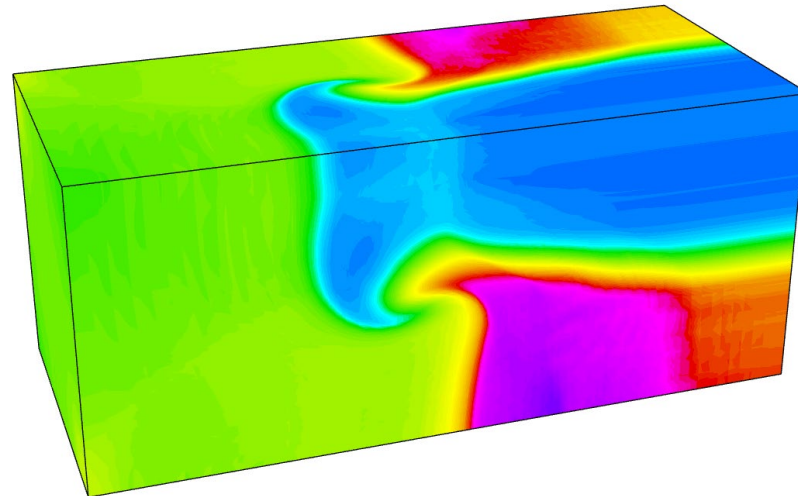
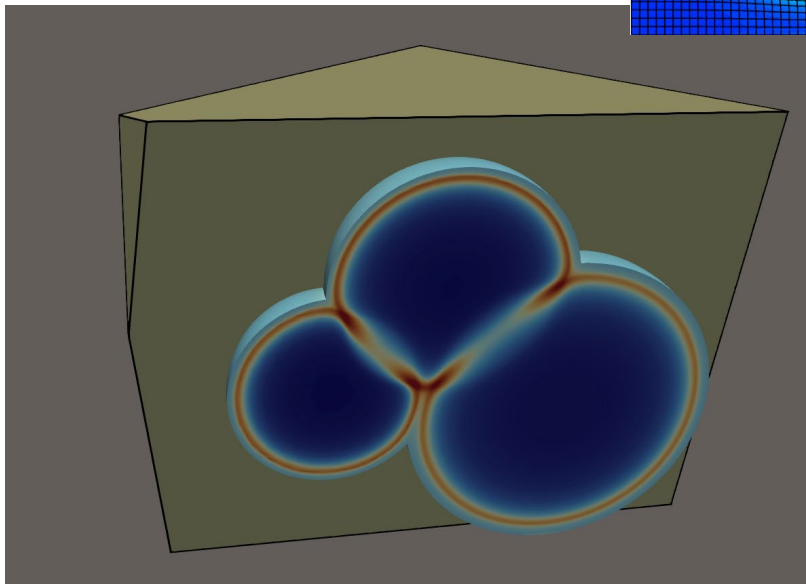
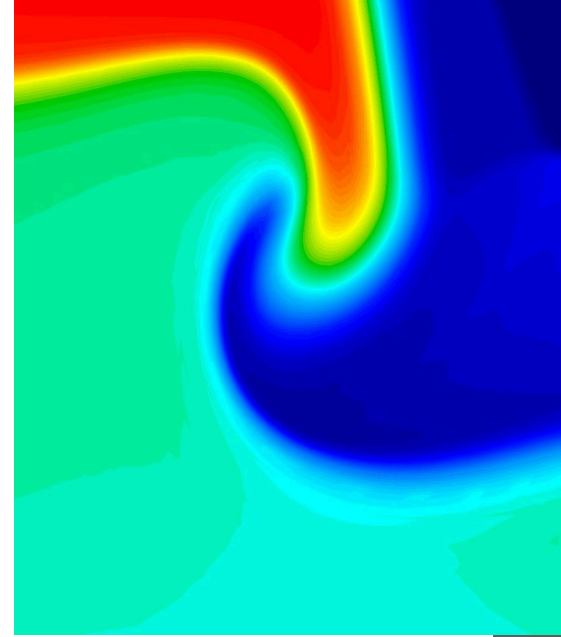
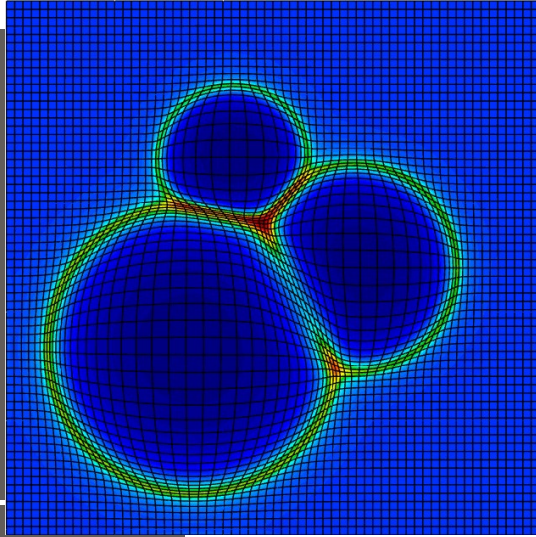
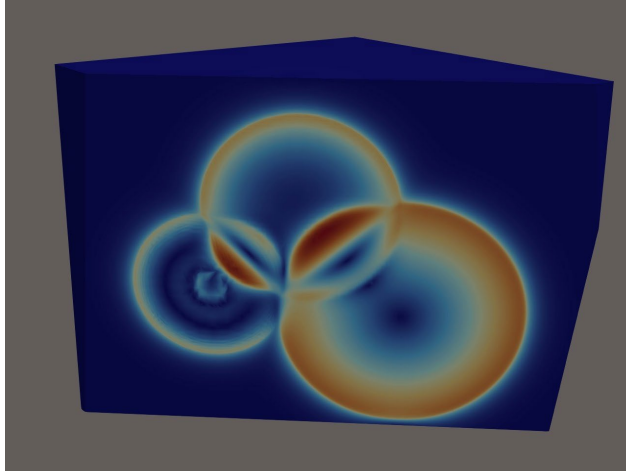
- IGR Does not widen the shear layer



Taylor Green has Second Order Convergence

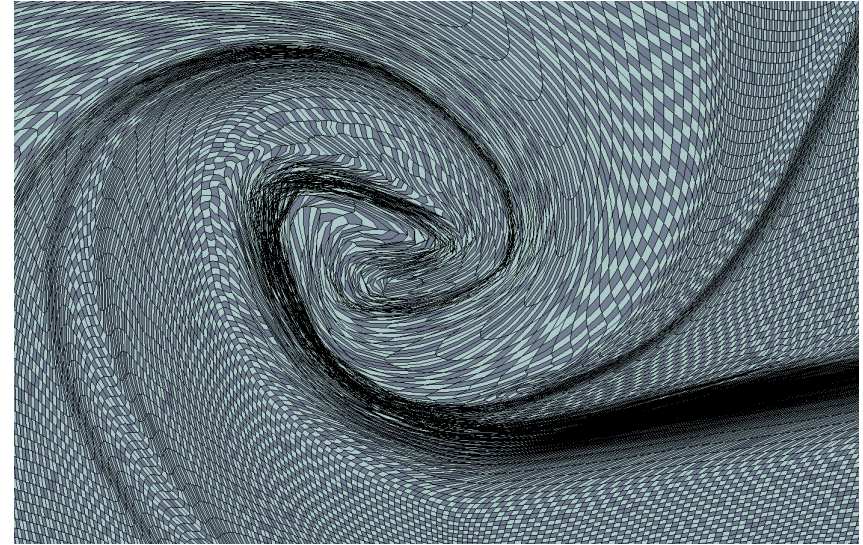
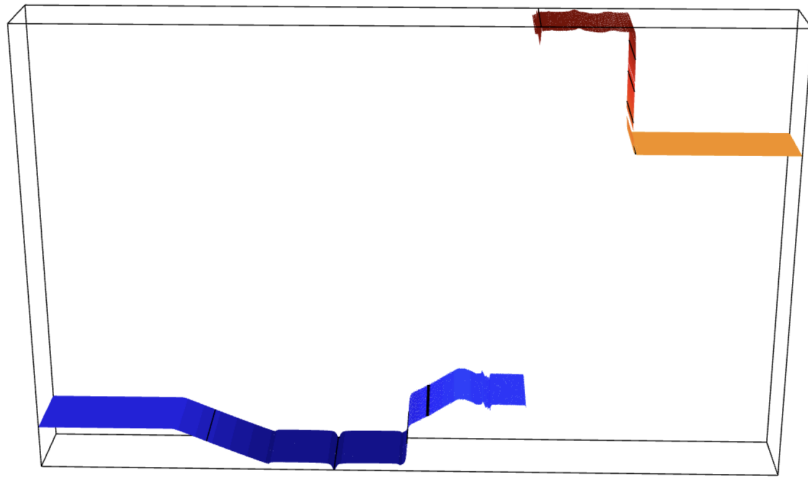


3d runs



Future Work

- We currently need to smooth kinematic velocity and pressure but can have discontinuous density and internal energy



- We will see if ALE methods can help with mesh tangling

Conclusion

Features of IGR:

- Prevents shocks without using viscosity
- Preserves coarse features at lower resolutions
- Larger time steps
- Greatly parallelizable
- Adds one extra elliptic solve per RK step

